Fiscal Year:	FY 2010	**Task Last Updated:**	FY 04/16/2010
PI Name:	Cavanagh, Peter R. Ph.D., D.Sc.		
Project Title:	A Quantitative Test of On-Orbit Exercise Countermeasures for Bone Demineralization Using a Bed Rest Analog-NNX08BA58G		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	206-221-2845
Organization Name:	University of Washington		
PI Address 1:	Department of Orthopaedics and Sports Medicine		
PI Address 2:	School of Medicine		
PI Web Page:			
City:	Seattle	**State:**	WA
Zip Code:	98195-6500	**Congressional District:**	7
Comments:	PI moved from Cleveland Clinic to University of Washington in June 2008 (8/08)		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	10/01/2008	**End Date:**	08/31/2011
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:	Meck, J@n	**Contact Phone:**	281-244-5405
Contact Email:	janice.v.meck@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Received funding augmentation/extension to 8/31/2011 (per NSSC info 2/2010) NOTE: Received NCE to 12/31/2010 per JSC (8/09) NOTE: Received NCE to 7/31/2009 (per NSSC info; added 8/09) NOTE: Received NCE to 4/30/2009; original end date was 12/31/2008 (per NSSC info; added 8/09)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Davis, Brian (Cleveland Clinic) Licata, Angelo (Cleveland Clinic) Ilaslan, Hakan (Cleveland Clinic)		
Grant/Contract No.:	NNX08BA58G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Bedrest is an excellent analog for the study of on-orbit musculoskeletal changes that occur during long-duration spaceflight. Our novel approach involves quantifying typical daily loads to the lower extremities in selected free-living volunteer subjects who are moderate exercisers and planning individual exercise prescriptions based on DLS theory. The specific aim of this project is to determine if individualized mechanical load replacement and supplementation during bedrest can serve as a countermeasure to losses in BMD, bone quality, and muscle atrophy during 12 weeks of bedrest.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The information obtained as a result of this study will provide further insight into the role of skeletal loading in bone loss during long-duration spaceflight, and knowledge gained from this research will thereby contribute to a better understanding of the importance of exercise for the development and maintenance of bone strength among humans living in Earth gravitational fields. Knowledge in this area is crucial to the treatment of bone disease for which exercise may or may not be an effective intervention.
Task Progress:	The total number of subjects who have completed the study at the Cleveland Clinic is now 11 (6 controls and 5 exercisers). During the last year, the administrative and regulatory procedures needed to transfer this study to the NASA Flight Analogs Research Unit (FARU) have been completed. The study has been successfully relocated to the NASA Flight Analogs Research Unit at the University of Texas Medical Branch in Galveston, TX for completion of the remaining subjects. It is anticipated that the first subjects in the Texas arm of the study will begin bedrest in early 2010. The project team has modified existing mathematical theories on daily load stimulus (DLS) incorporating new theories on saturation and recovery of the osteogenic potential of bone with distributed mechanical loading and the effects of standing. This revised algorithm has been dubbed the enhanced DLS (EDLS). Extensive MATLAB code has been written to allow users to analyze complete sets of in-shoe force data and calculate EDLS. The EDLS is being used to formulate the individualized exercise prescriptions used by the exercising subjects. The EDLS algorithm has been documented and published in the journal "Aviation, Space, and Environmental Medicine."
Bibliography Type:	Description: (Last Updated: 03/08/2018)
Abstracts for Journals and Proceedings	Cavanagh PR, Licata AA, Rice AJ, Gopalakrishnan R, Novotny SC, Kuklis MM, Davis BL, Genc KO, Englehaupt RK, Rizk RY, Ilaslan H. "Exercise as a countermeasure to bone demineralization during 12-weeks of bedrest." NASA Human Research Program Investigators' Workshop, League City, TX, February 2-4, 2009. Proceedings from NASA Human Research Program Investigators' Workshop, League City, TX, February 204, 2009. , Feb-2009
Abstracts for Journals and Proceedings	Genc KO, Humphreys BT, Rice AJ, Englehaupt RK, Novotny SC, Gopalakrishnan R, Ilaslan H, Licata AA, Cavanagh PR. "Validating the enhanced daily load stimulus model using the bedrest analog of spaceflight." 33rd Annual Meeting of the American Society of Biomechanics, University Park, PA, August 26-29, 2009. Proceedings from the 33rd Annual Meeting of the American Society of Biomechanics, University Park, PA, August 26-29, 2009. , Aug-2009
Abstracts for Journals and Proceedings	Cavanagh PR, Rice AJ, Novotny SC, Hanson AM, Gopalakrishnan R, Kuklis MM, Davis BL, Genc KO, Englehaupt RK, Rizk RY, Ilaslan H, Licata A. "Exercise as a countermeasure to bone demineralization during 12-weeks of bedrest." NASA Human Research Program Investigators' Workshop, Houston, TX, February 3–5, 2010. Proceedings from the NASA Human Research Program Investigators' Workshop, Houston, TX, February 3–5, 2010. , Feb-2010
Articles in Peer-reviewed Journals	Genc KO, Humphreys BT, Cavanagh PR. "Enhanced daily load stimulus to bone in spaceflight and on earth." Aviat Space Environ Med. 2009 Nov;80(11):919-26. PubMed PMID: 19911514 , Nov-2009